WearUS: A Unified Framework for Wearable A-Mode Ultrasound Pretraining and Downstream Transfer

Ninth Weekly Report

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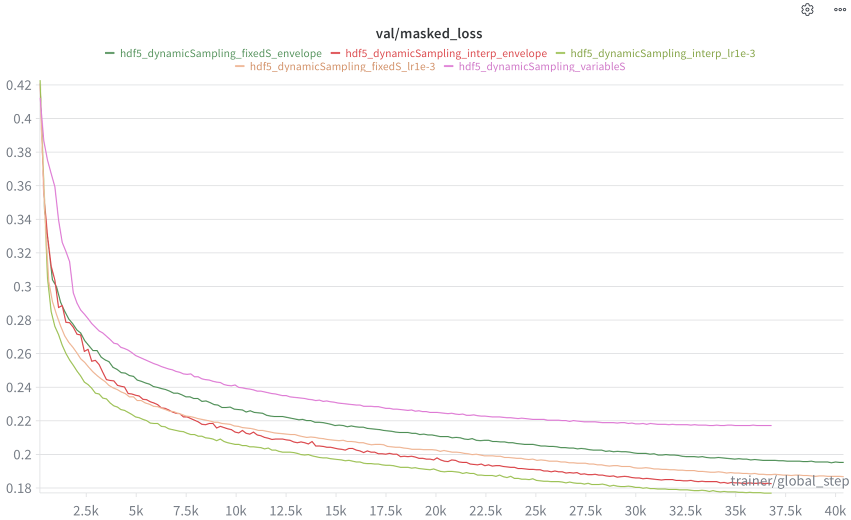
Ablation Study: Sampling & Preprocessing Results

Experiments Conducted

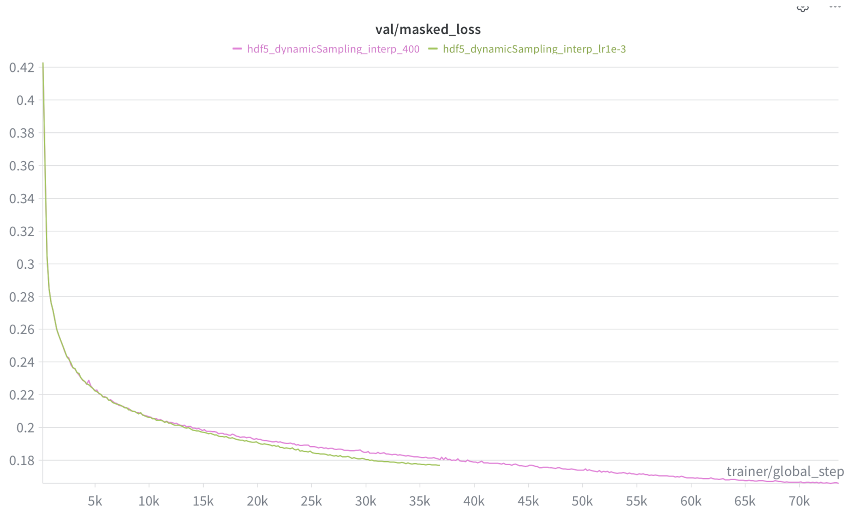
Sampling Strategy	**Mode**	**Preprocessing**	**min_loss**
Naive	FixedS	raw	0.2147
Naive	interpolation	raw	0.2044
Dynamic	FixedS	raw	0.1869
Dynamic	**interpolation**	**raw**	**0.1771**
Dynamic	VariableS	raw	0.1910
Dynamic	FixedS	envelope	0.1954
Dynamic	interpolation	envelope	0.1829

Loss Function Trends



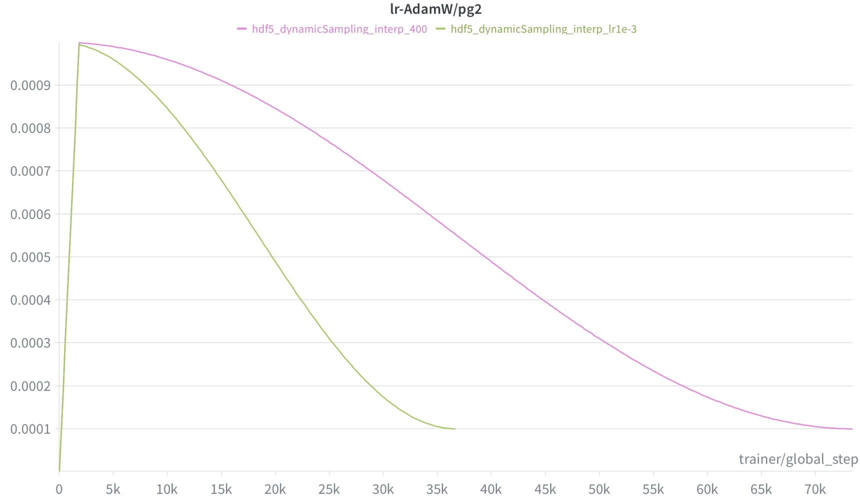
Comparison of training loss convergence across different sampling and preprocessing configurations.

Training Duration: 200 vs. 400 Epochs



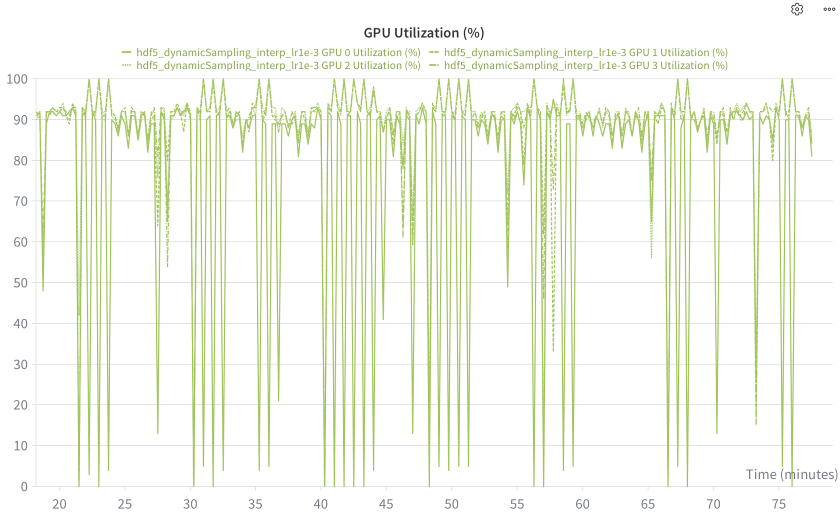
min_loss on 200 epochs: 0.1771 — min_loss on 400 epochs: 0.1659

Learning Rate Scheduling



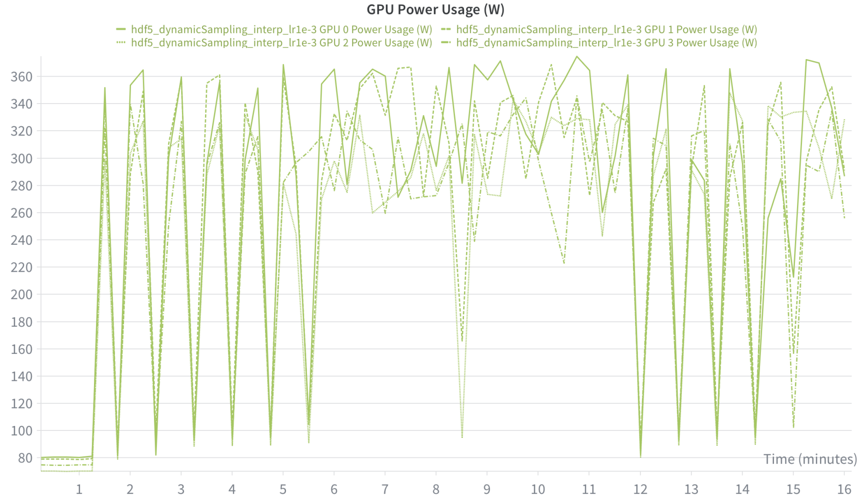
Policy: Cosine Scheduler

GPU Utilization



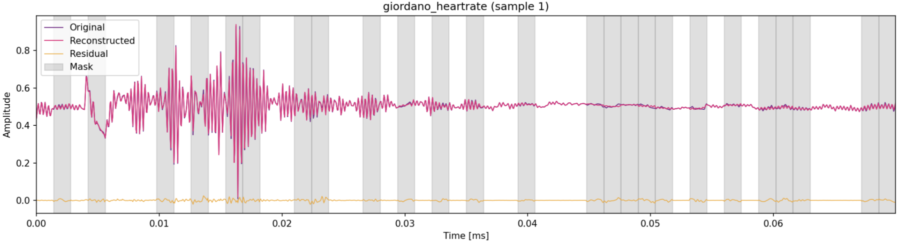
Hardware Performance Monitoring

Power Consumption



System Energy Footprint Monitoring

Reconstruction Example



Visual validation: Original signal vs. Model reconstruction